

Artificial Intelligence

An Advanced Course About Foundations



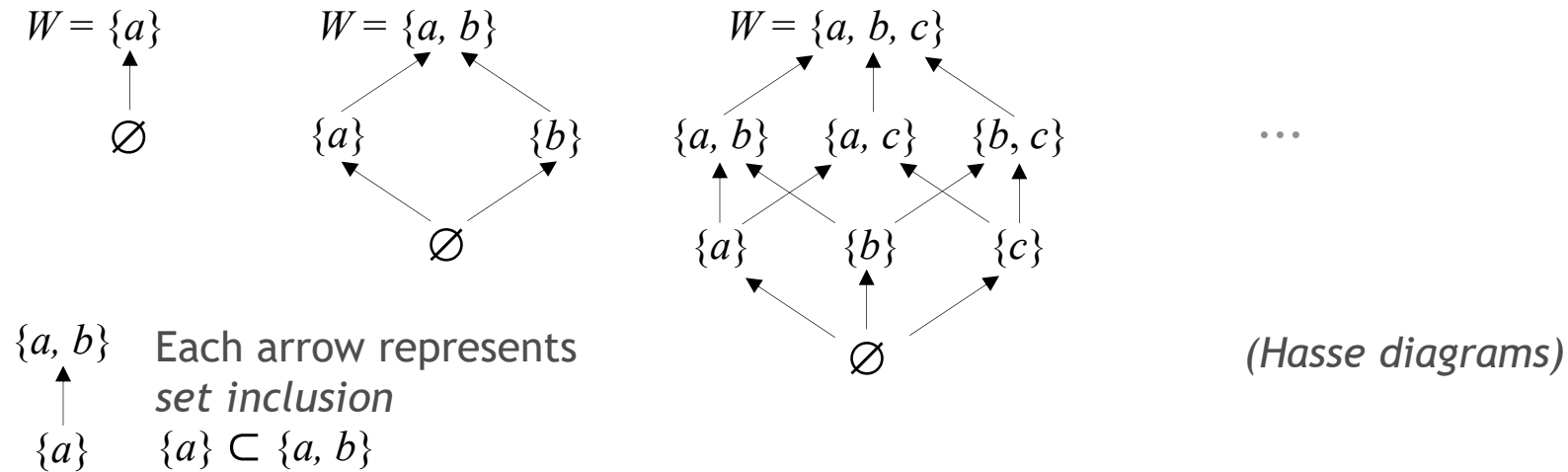
Propositional Logic

Marco Piastra

Prologue: Boolean Algebra(s)

Boolean algebras by examples

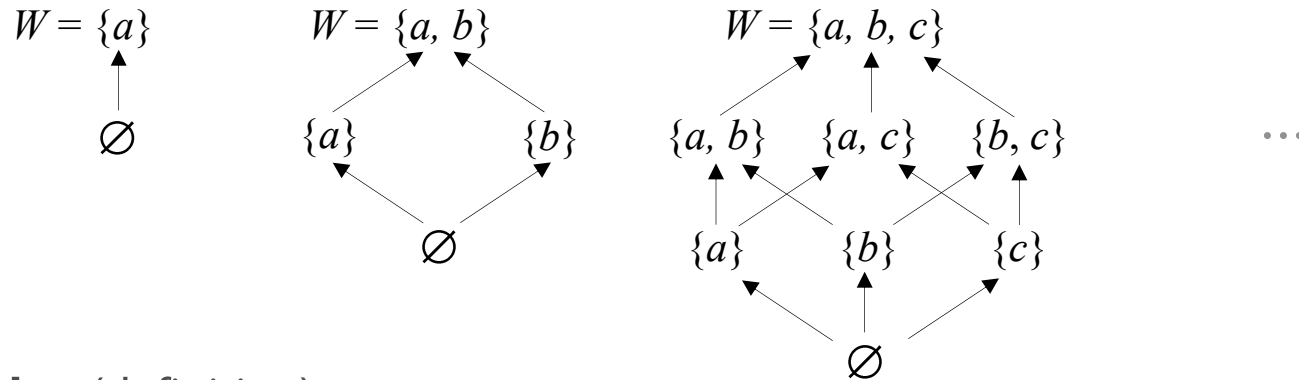
Consider a *finite* set of objects W and construct, in a *bottom-up fashion*, the collection Σ of all possible subsets of W



Collections like Σ above are also called the **power set** of W which is the collection of all possible subsets of W , also denoted as 2^W

Boolean algebras by examples

Consider a *finite* set of objects W and construct, in a *bottom-up fashion*, the collection Σ of all possible subsets of W



Boolean algebra (definition)

Any non-empty collection of subsets Σ of a set W such that:

- 1) $\emptyset \in \Sigma$
- 2) $A, B \in \Sigma \implies A \cup B \in \Sigma$
- 3) $A \in \Sigma \implies A^c \in \Sigma$

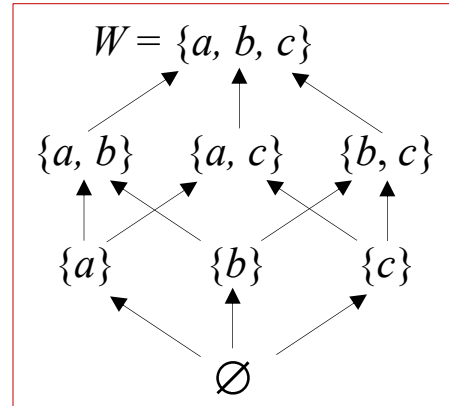
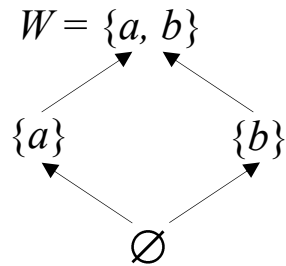
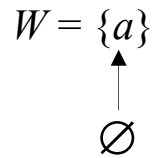
$A^c := W - A$ (the complement of A with respect to W)

Corollaries:

- The set W belongs to any Boolean algebra generated on W
- Σ is closed under *intersection* $\cap$

Boolean algebras by examples

Consider a *finite* set of objects W and construct, in a *bottom-up* fashion, the collection Σ of all possible subsets of W



...

De Morgan's laws

Checking properties of a **Boolean algebra**

For any of the (finite) structures above properties can be verified exhaustively

$$(A \cup B)^c = A^c \cap B^c$$

$$\begin{aligned} A &= \{b\} \\ B &= \{b, c\} \\ A \cup B &= \{b, c\} \\ (A \cup B)^c &= \{a\} \\ A^c &= \{a, c\} \\ B^c &= \{a\} \\ A^c \cap B^c &= \{a\} \end{aligned}$$

These sets
are identical



$$(A \cap B)^c = A^c \cup B^c$$

$$\begin{aligned} A &= \{b\} \\ B &= \{b, c\} \\ A \cap B &= \{b\} \\ (A \cap B)^c &= \{a, c\} \\ A^c &= \{a, c\} \\ B^c &= \{a\} \\ A^c \cup B^c &= \{a, c\} \end{aligned}$$

These sets
are identical



Which Boolean algebra for logic?

*Given that all boolean algebras share the same properties
we can adopt the simplest one as reference: the one based on $\Sigma := \{W, \emptyset\}$*

This is a two-valued algebra: $\{\text{nothing}, \text{everything}\}$ or $\{\text{false}, \text{true}\}$ or $\{\perp, \top\}$ or $\{0, 1\}$

- Algebraic structure

$< \{0,1\}, \text{OR}, \text{AND}, \text{NOT} >$

- Boolean functions and *truth tables*

Most generic type of *boolean functions*: $f: \{0, 1\}^n \rightarrow \{0, 1\}$

AND, OR and NOT are boolean functions, defined extensionally via *truth tables*

A	B	OR
0	0	0
0	1	1
1	0	1
1	1	1

A	B	AND
0	0	0
0	1	0
1	0	0
1	1	1

A	NOT
0	1
1	0

Composite functions

Truth tables can be defined also for composite functions

For example, to verify logical laws

These columns are identical

A	B	NOT A	NOT B	A OR B	NOT(A OR B)	NOT A AND NOT B
0	0	1	1	0	1	1
0	1	1	0	1	0	0
1	0	0	1	1	0	0
1	1	0	0	1	0	0

De Morgan's laws

These columns are identical

A	B	NOT A	NOT B	A AND B	NOT(A AND B)	NOT A OR NOT B
0	0	1	1	0	1	1
0	1	1	0	0	1	1
1	0	0	1	0	1	1
1	1	0	0	1	0	0

Adequate basis

- How many boolean functions do we need to define *any* boolean function?

$\uparrow$ 2^n rows $\downarrow$	A_1	A_2	...	A_n	$f(A_1, A_2, \dots, A_n)$
	0	0	...	0	f_1
	0	0	...	1	f_2
	...	...	...	...	...
	...	...	...	...	...
	1	1	...	1	f_{2^n}

Just OR, AND and NOT : any other function can be expressed as composite function

In the generic *truth table* above:

- For each row j where $f_j = 1$, create a Boolean expression composing by AND the n input variables, taking either A_i , when the i -th value is 1, or NOT A_i when i -th value is 0
- Compose with OR all expressions obtained in the way above

Other adequate bases

Also $\{\text{OR}, \text{NOT}\}$ o $\{\text{AND}, \text{NOT}\}$ are adequate bases

An adequate basis can be obtained by just one 'ad hoc' function: NOR or NAND

A	B	$A \text{ NOR } B$
0	0	1
0	1	0
1	0	0
1	1	0

A	B	$A \text{ NAND } B$
0	0	1
0	1	1
1	0	1
1	1	0

- Two remarkable functions: *implication* and *equivalence*

A	B	$A \text{ IMP } B$
0	0	1
0	1	1
1	0	0
1	1	1

A	B	$A \text{ EQU } B$
0	0	1
0	1	0
1	0	0
1	1	1

Identities:

$$A \text{ IMP } B = (\text{NOT } A) \text{ OR } B$$

$$A \text{ EQU } B = (A \text{ IMP } B) \text{ AND } (B \text{ IMP } A)$$

In passing, logicians prefer the adequate basis $\{\text{IMP}, \text{NOT}\}$

Language and Semantics, *Possible Worlds*

Propositional logic: the project

The simplest of 'classical' logics

■ Propositions

We consider simple **propositions** which state something that could be either true or false (in a context)

"Today is Friday"

"Turkeys are birds with feathers"

"Man is a featherless biped"

■ Formal *language*

A precise and formal language whose **atoms** are *propositions*
(no intention to represent the internal structure of *propositions*)

Atoms will be composed in complex formulae via a set of *syntactic* rules

■ Formal *semantics*

A class of *formal* structures, each representing a **possible world** or a possible 'state of things'

<This classroom right now>

<My uncle's farm several years ago>

<Ancient Greece at the time of Aristotle's birth>

The class of propositional semantic structures

Possible world

A structure $\langle \{0,1\}, \Sigma, v \rangle$

- $\{0,1\}$ are the *truth values*
- Σ is the **signature** of the formal language: a set of propositional symbol
- v is a *function* : $\Sigma \rightarrow \{0,1\}$ assigning a truth value to every symbol in Σ

Propositional symbols (*signature*)

Each symbol in Σ represents an actual *proposition* (in natural language)

By convention, we use the symbols $A, B, C, D, \dots$

Caution: Σ needs not be *finite*...

Possible worlds

The class of structures contains all possible worlds:

$\langle \{0,1\}, \Sigma, v \rangle$

$\langle \{0,1\}, \Sigma, v' \rangle$

$\langle \{0,1\}, \Sigma, v'' \rangle$

...

All structure in a class share the same Σ and $\{0,1\}$

The functions v are *different*: the assignment of truth values varies, depending on the possible world

Formal *language*

■ Propositional language L_P

- A set Σ of propositional symbols: $\Sigma = \{A, B, C, \dots\}$
- Two (primary) **logical connectives**: $\neg, \rightarrow$
- Three (derived) **logical connectives**: $\wedge, \vee, \leftrightarrow$
- Parenthesis: $(,)$ (there are no *precedence rules* in this language)

■ Well-formed formulae (**wff**)

Defined via a set of syntactic rules:

The set of all the **wff** of L_P is denoted as $\text{wff}(L_P)$

$$A \in \Sigma \Rightarrow A \in \text{wff}(L_P)$$

$$\varphi \in \text{wff}(L_P) \Rightarrow (\neg\varphi) \in \text{wff}(L_P)$$

$$\varphi, \psi \in \text{wff}(L_P) \Rightarrow (\varphi \rightarrow \psi) \in \text{wff}(L_P)$$

$$\varphi, \psi \in \text{wff}(L_P) \Rightarrow (\varphi \vee \psi) \in \text{wff}(L_P)$$

$$\varphi, \psi \in \text{wff}(L_P) \Rightarrow (\varphi \wedge \psi) \in \text{wff}(L_P)$$

$$\varphi, \psi \in \text{wff}(L_P) \Rightarrow (\varphi \leftrightarrow \psi) \in \text{wff}(L_P)$$

Formal *semantics*: interpretations

■ Compositional (*truth-functional*) semantics for wff

Given a possible world $\langle \{0,1\}, \Sigma, \nu \rangle$ the function $\nu : \Sigma \rightarrow \{0,1\}$ can be extended by associating a *boolean function* to each *connective*:

$$\begin{aligned}\nu(\neg\varphi) &= \text{NOT}(\nu(\varphi)) \\ \nu(\varphi \wedge \psi) &= \text{AND}(\nu(\varphi), \nu(\psi)) \\ \nu(\varphi \vee \psi) &= \text{OR}(\nu(\varphi), \nu(\psi)) \\ \nu(\varphi \rightarrow \psi) &= \text{OR}(\text{NOT}(\nu(\varphi)), \nu(\psi)) \quad (\text{also } \text{IMP}(\nu(\varphi), \nu(\psi)) \text{ }) \\ \nu(\varphi \leftrightarrow \psi) &= \text{AND}(\text{OR}(\text{NOT}(\nu(\varphi)), \nu(\psi)), \text{OR}(\text{NOT}(\nu(\psi)), \nu(\varphi)))\end{aligned}$$

■ Interpretations

Function ν (extended as above) assigns a truth value to each $\varphi \in \text{wff}(L_P)$

$$\nu : \text{wff}(L_P) \rightarrow \{0,1\}$$

Then ν is said to be an *interpretation* of L_P

Note that the truth value of any $\varphi \in \text{wff}(L_P)$ is univocally determined by the values assigned by ν to each symbol in the *signature* Σ

Formalism: object *language* and *metalanguage*

- The **object language** is L_P

The formal language of logic

It contains **all** $\varphi \in \text{wff}(L_P)$

- **Meta-language**

The formalism for defining the properties of the object language and the logic

Small Greek letters ($\alpha, \beta, \chi, \varphi, \psi, \dots$) will be used to denote formulae (wff)

Capital Greek letters ($\Gamma, \Delta, \dots$) will be used to denote sets of formulae

Satisfaction, logical consequence (see after): $\models$

Derivability (see after): $\vdash$

“if and only if” : “iff”

Implication, equivalence (in general): $\Rightarrow, \Leftrightarrow$

Entailment

About formulae and their hidden relations

■ Hypothesis:

$$\varphi_1 = B \vee D \vee \neg(A \wedge C)$$

"Sally likes Harry" OR "Harry is happy"
OR NOT ("Harry is human" AND "Harry is a featherless biped")

$$\varphi_2 = B \vee C$$

"Sally likes Harry" OR "Harry is a featherless biped"

$$\varphi_3 = A \vee D$$

"Harry is human" OR "Harry is happy"

$$\varphi_4 = \neg B$$

NOT "Sally likes Harry"

■ Thesis:

$$\psi = D$$

"Harry is happy"

Is there any *logical relation*
between hypothesis and thesis?

And among the propositions
in the hypothesis?

Entailment

The overall truth table
for the wff in the example

$$\varphi_1 = B \vee D \vee \neg(A \wedge C)$$

$$\varphi_2 = B \vee C$$

$$\varphi_3 = A \vee D$$

$$\varphi_4 = \neg B$$

$$\psi = D$$

Entailment

$$\{\varphi_1, \varphi_2, \varphi_3, \varphi_4\} \models \psi$$

Notation!

There is entailment when
all the *possible worlds* that *satisfy* $\{\varphi_1, \varphi_2, \varphi_3, \varphi_4\}$
satisfy ψ as well

A	B	C	D	φ_1	φ_2	φ_3	φ_4	ψ
0	0	0	0	1	0	0	1	0
0	0	0	1	1	0	1	1	1
0	0	1	0	1	1	0	1	0
0	0	1	1	1	1	1	1	1
0	1	0	0	1	1	0	0	0
0	1	0	1	1	1	1	0	1
0	1	1	0	1	1	0	0	0
0	1	1	1	1	1	1	0	1
1	0	0	0	1	0	1	1	0
1	0	0	1	1	0	1	1	1
1	0	1	0	0	1	1	1	0
1	0	1	1	1	1	1	1	1
1	1	0	0	1	1	1	0	0
1	1	0	1	1	1	1	0	1
1	1	1	0	1	1	1	0	0
1	1	1	1	1	1	1	0	1

A set of wff

One wff

$$\Gamma \models \varphi$$

*There is entailment iff
every world that satisfies Γ
also satisfies φ*

Satisfaction, models

■ Possible worlds and *truth tables*

Examples: $\varphi = (A \vee B) \wedge C$

Different rows, different groups of worlds

All rows, all possible worlds

Caution: in each possible world

every $\varphi \in \text{wff}(L_P)$ has a truth value

so a row in a table is not a single world, per se

A	B	C	$A \vee B$	$(A \vee B) \wedge C$
0	0	0	0	0
0	0	1	0	0
0	1	0	1	0
0	1	1	1	1
1	0	0	1	0
1	0	1	1	1
1	1	0	1	0
1	1	1	1	1

A possible world **satisfies** a wff φ iff $v(\varphi) = 1$

We also write $\langle \{0,1\}, \Sigma, v \rangle \models \varphi$

In the truth table above, the rows that satisfy φ are in gray

Such possible world w is also said to be a **model** of φ

By extension, a possible world *satisfies* (i.e. is *model* of) a set of wff $\Gamma = \{\varphi_1, \varphi_2, \dots, \varphi_n\}$

iff w *satisfies* (i.e. is *model* of) each of its wff $\varphi_1, \varphi_2, \dots, \varphi_n$

Tautologies, contradictions

■ A tautology

Is a (propositional) wff
that is always satisfied

It is also said to be **valid**

Any wff of the type $\varphi \vee \neg\varphi$
is a tautology

■ A contradiction

Is a (propositional) wff,
that cannot be satisfied

Any wff of the type $\varphi \wedge \neg\varphi$
is a contradiction

Notes:

- Not all wff are either *tautologies* or contradictions
- If φ is a *tautology* then $\neg\varphi$ is a *contradiction* and vice-versa

A	$A \wedge \neg A$	$A \vee \neg A$
0	0	1
1	0	1

A	B	$(\neg A \vee B) \vee (\neg B \vee A)$
0	0	1
0	1	1
1	0	1
1	1	1

A	B	$\neg((\neg A \vee B) \vee (\neg B \vee A))$
0	0	0
0	1	0
1	0	0
1	1	0

Formulae and subsets

- Consider the set W of all possible worlds

Each wff φ of L_P corresponds to a **subset** of W

The subset of all possible worlds that *satisfy* it

In other words, φ corresponds to $\{w : w \models \varphi\}$

The corresponding subset may be empty (i.e. if φ is a contradiction)
or it may coincide with W (i.e if φ is a tautology)

The set of all
possible worlds



Formulae and subsets

- Consider the set W of all possible worlds

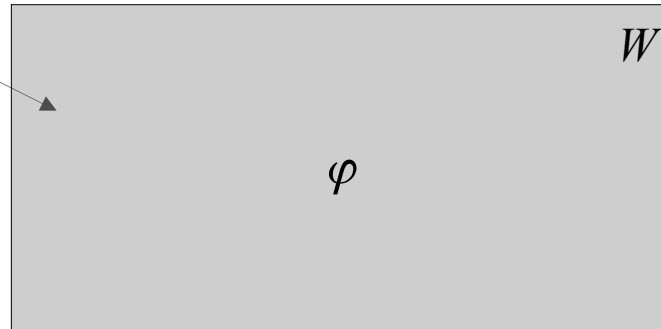
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The set of all
possible worlds



“ φ is a tautology”

“any possible world in W
is a *model* of φ ”

“ φ is *valid*”

Furthermore:

“ φ is *satisfiable*”

“ φ is not *falsifiable*”

Formulae and subsets

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The set of all
possible worlds



“ φ is a contradiction”

“none of the possible worlds in W
is a *model* of φ ”

“ φ is not *valid*”

Furthermore:

“ φ is not *satisfiable*”

“ φ is *falsifiable*”

Formulae and subsets

- Consider the set W of all possible worlds

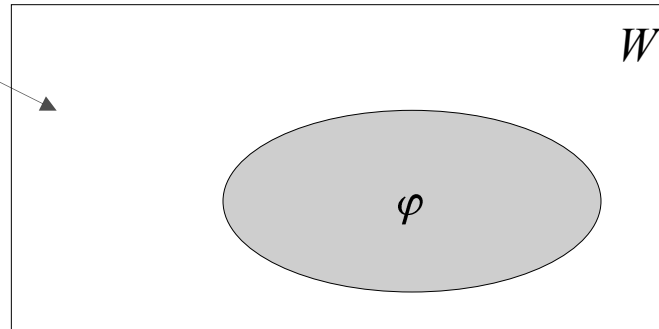
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The corresponding subset may be empty (i.e. if φ is a contradiction)
or it may coincide with W (i.e if φ is a tautology)

The set of all
possible worlds



“ φ is neither a contradiction
nor a tautology”

“some possible worlds in W
are *model* of φ , others are not”

“ φ is not (logically) **valid**”

Furthermore:

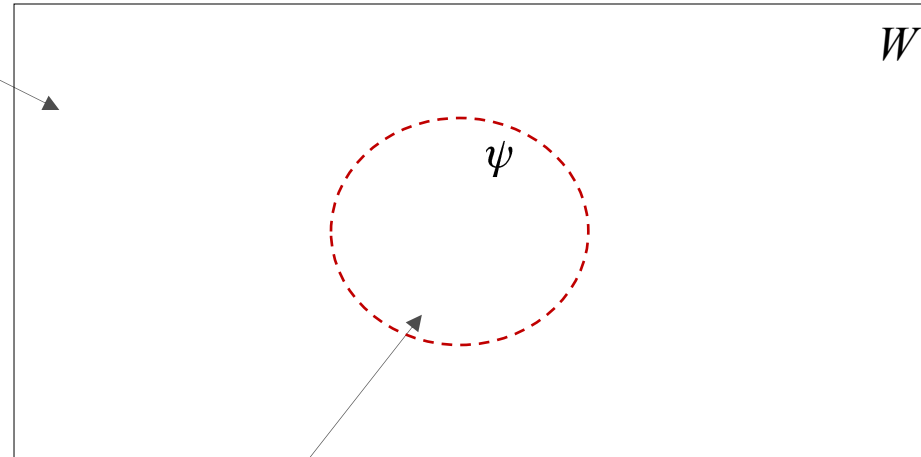
“ φ is **satisfiable**”

“ φ is **falsifiable**”

Formulae, subsets and entailment

Another view over the “Harry is happy” example

All possible worlds



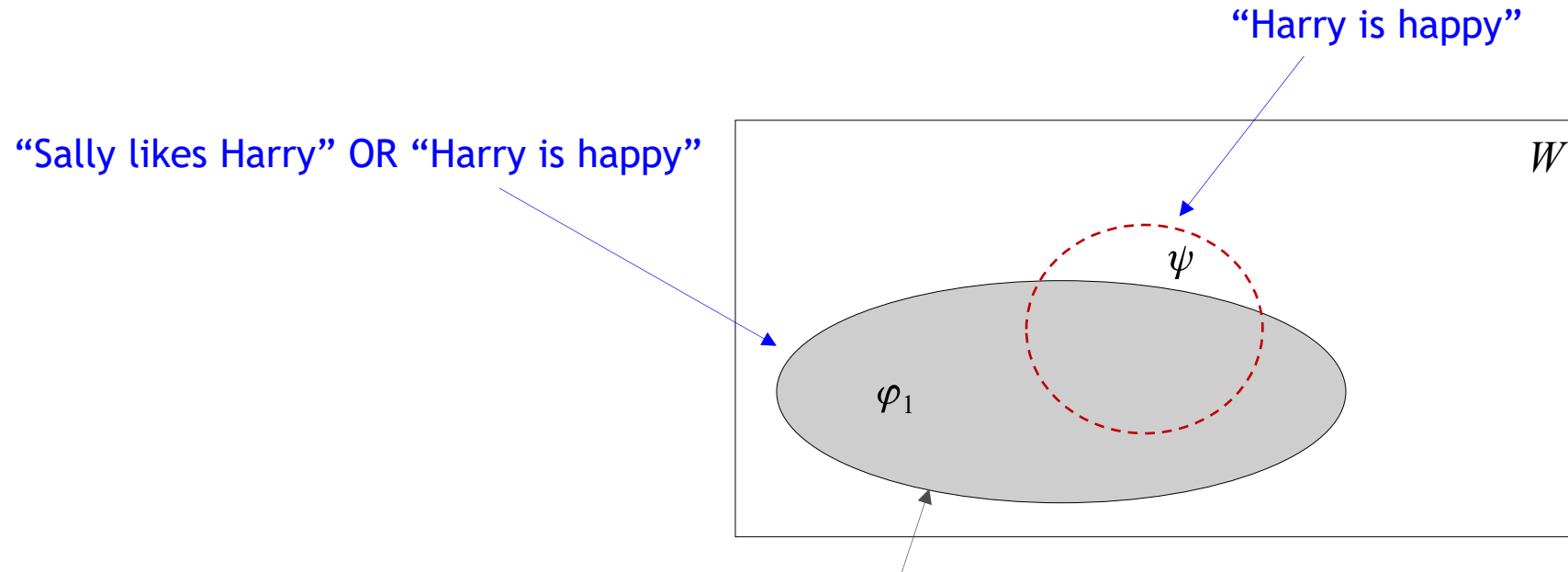
“All possible worlds that are *satisfy* ψ ”

“All possible worlds that are *models* of ψ ”

Say it with sets:
the example revisited

Formulae, subsets and entailment

Another view over the “Harry is happy” example



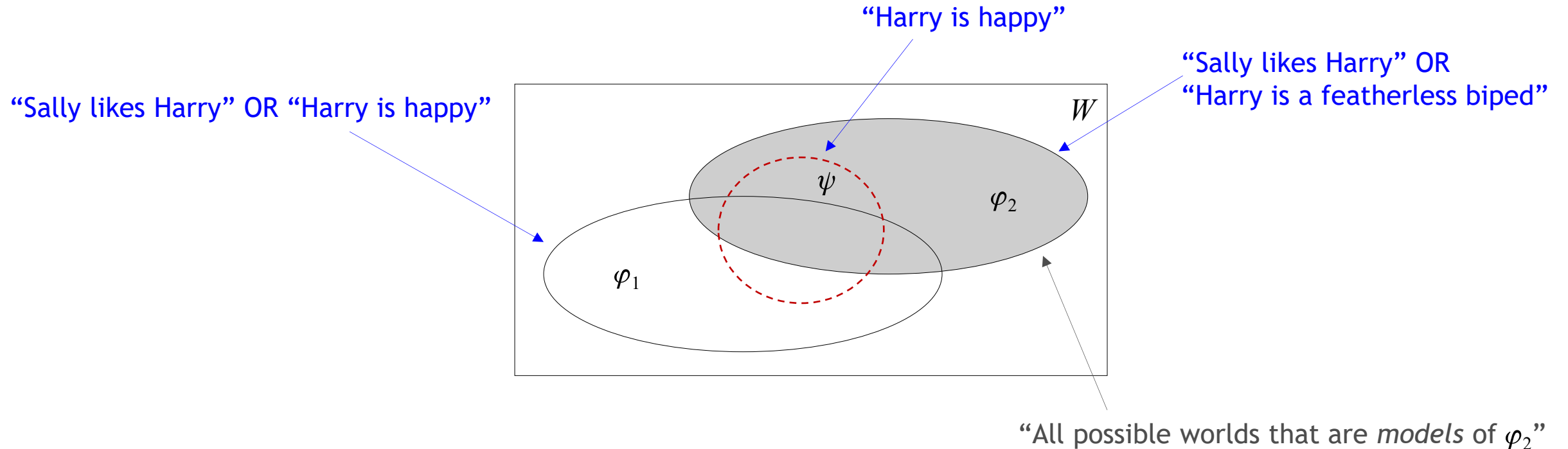
“All possible worlds that are *models* of φ_1 ”

$\{\varphi_1\} \not\models \psi$

because the set of models for $\{\varphi_1\}$
is not contained in the set of models of ψ

Formulae, subsets and entailment

Another view over the “Harry is happy” example

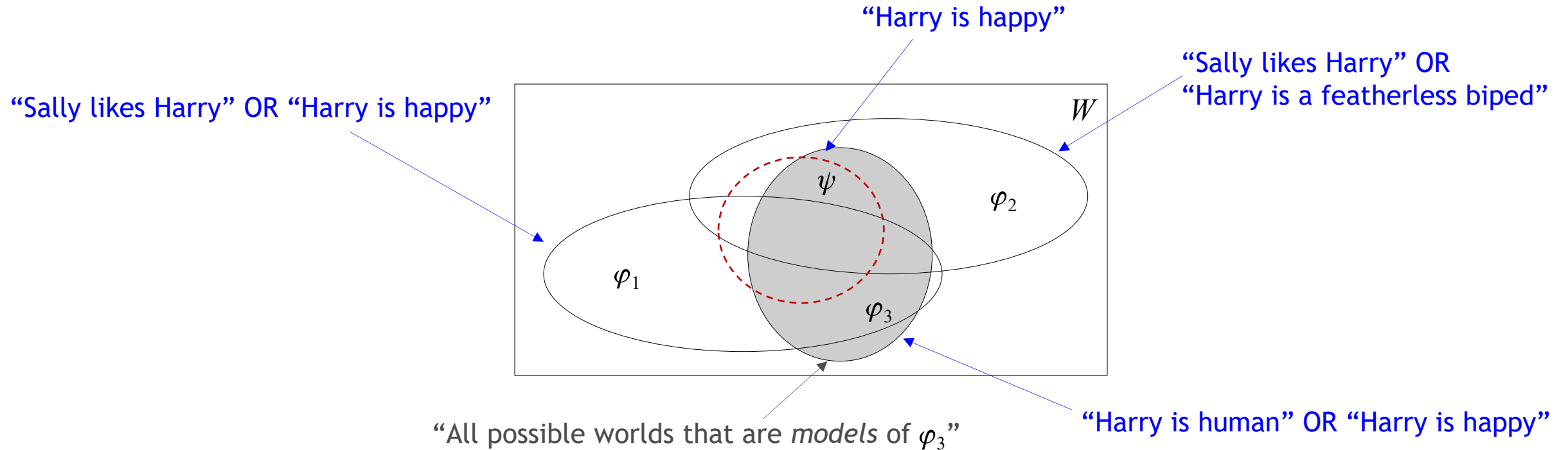


$\{\varphi_1, \varphi_2\} \not\models \psi$

because the set of models of $\{\varphi_1, \varphi_2\}$ (i.e. the *intersection* of the two subsets) is not contained in the set of models of ψ

Formulae, subsets and entailment

Another view over the “Harry is happy” example

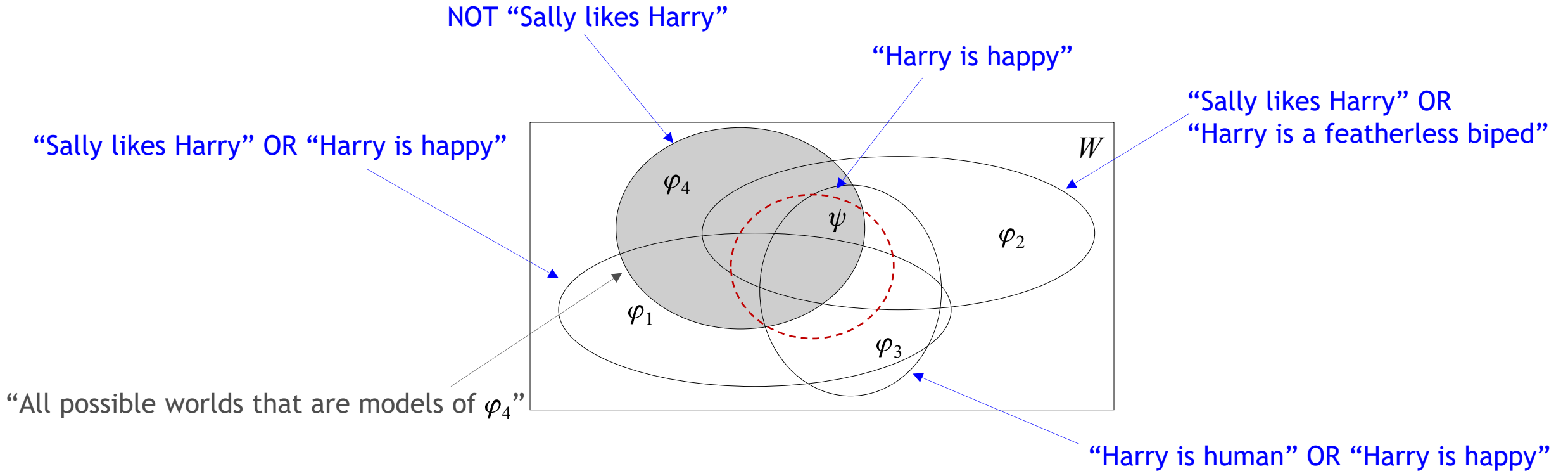


$\{\varphi_1, \varphi_2, \varphi_3\} \not\models \psi$

because the set of models of $\{\varphi_1, \varphi_2, \varphi_3\}$
is not contained in the set of models of ψ

Formulae, subsets and entailment

Another view over the “Harry is happy” example

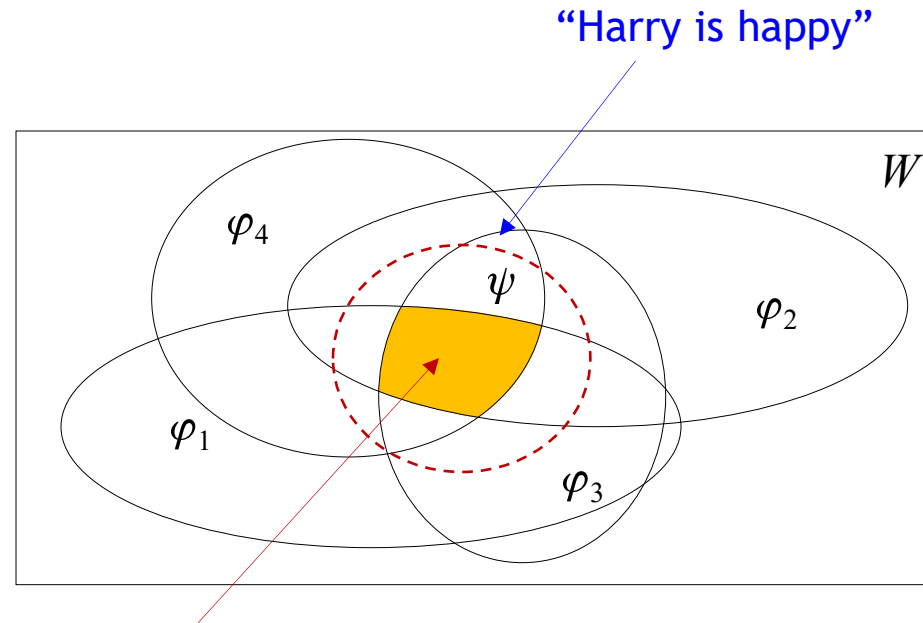


$$\{\varphi_1, \varphi_2, \varphi_3, \varphi_4\} \models \psi$$

Because the set of models for $\{\varphi_1, \varphi_2, \varphi_3, \varphi_4\}$
is contained in the set of models of ψ

Formulae, subsets and entailment

Another view over the “Harry is happy” example



“All possible worlds that are models for $\{\varphi_1, \varphi_2, \varphi_3, \varphi_4\}$ ”

$$\{\varphi_1, \varphi_2, \varphi_3, \varphi_4\} \models \psi$$

Because the set of models for $\{\varphi_1, \varphi_2, \varphi_3, \varphi_4\}$
is contained in the set of models of ψ

In the case of the example,
all the wff $\varphi_1, \varphi_2, \varphi_3, \varphi_4$
are needed for the relation
of *entailment* to hold

A set of wff

One wff

$$\Gamma \models \varphi$$

*There is entailment iff
every world that satisfies Γ
also satisfies φ*

Further Properties

Symmetric entailment = logical equivalence

■ Equivalence

Let φ and ψ be wff such that:

$$\varphi \models \psi \text{ e } \psi \models \varphi$$

The two wff are also said to be **logically equivalent**

In symbols: $\varphi \equiv \psi$

■ Substitutability

Two wff are equivalent iff they have *the same models*

In terms of entailment, equivalent wff are substitutable
(even as sub-formulae)

In the example: $\{\varphi_1, \varphi_2, \varphi_3, \varphi_4\} \models \psi$

$$\varphi_1 = B \vee D \vee \neg(A \wedge C)$$

$$\varphi_2 = B \vee C$$

$$\varphi_3 = A \vee D$$

$$\varphi_4 = \neg B$$

$$\psi = D$$

$$\varphi_1 = B \vee D \vee (A \rightarrow \neg C)$$

$$\varphi_2 = B \vee C$$

$$\varphi_3 = \neg A \rightarrow D$$

$$\varphi_4 = \neg B$$

$$\psi = D$$

Implication and Inference Schemas

The wff of the problem can be re-written using equivalent expressions:
(using the basis $\{\rightarrow, \neg\}$)

$$\varphi_1 = C \rightarrow (\neg B \rightarrow (A \rightarrow D))$$

$$\varphi_2 = \neg B \rightarrow C$$

$$\varphi_3 = \neg A \rightarrow D$$

$$\varphi_4 = \neg B$$

$$\psi = D$$

$$\varphi_1 = B \vee D \vee \neg(A \wedge C)$$

$$\varphi_2 = B \vee C$$

$$\varphi_3 = A \vee D$$

$$\varphi_4 = \neg B$$

$$\psi = D$$

- Some ***inference schemas*** are *valid* in terms of *entailment*:

$$\frac{\varphi \rightarrow \psi \quad \varphi}{\psi}$$

It can be verified that (*see the truth table of implication*):

$$\varphi \rightarrow \psi, \varphi \models \psi$$

Analogously:

$$\varphi \rightarrow \psi, \neg\psi \models \neg\varphi$$

Modern formal logic: fundamentals

- **Formal language (*symbolic*)**

- A set of symbols, not necessarily *finite*

- Syntactic rules for composite formulae (wff)

- **Formal semantics**

- For each formal language, a *class* of structures (i.e. a class of *possible worlds*)

- In each possible world, every wff in the language is assigned a *value*

- In classical propositional logic, the set of values is the simplest: $\{1, 0\}$

- **Satisfaction, entailment**

- A wff is *satisfied* in a possible world if it is true in that possible world

- In classical propositional logic, iff the wff has value 1 in that world

- (Caution: the definition of *satisfaction* will become definitely more complex with *first order logic*)

Entailment is a relation between a set of wff and a wff

- This relation holds when all possible worlds satisfying the set also satisfy the wff

Properties of entailment (classical logic)

■ Compactness

Consider a set of wff Γ (not necessarily *finite*)

$\Gamma \models \varphi \Rightarrow$ There exist a finite subset $\Sigma \subseteq \Gamma$ such that $\Sigma \models \varphi$

(This follows from *compositionality*, see textbook for a proof)

■ Monotonicity

For any Γ and Δ , if $\Gamma \models \varphi$ then $\Gamma \cup \Delta \models \varphi$

In fact, any entailment relation between φ and Γ remains valid even if Γ grows larger

■ Transitivity

If for all $\varphi \in \Sigma$ we have $\Gamma \models \varphi$, then if $\Sigma \models \psi$ then $\Gamma \models \psi$

If Γ entails any φ in Σ , then any ψ entailed by Σ is also entailed by Γ

■ *Ex absurdo ...*

$\{\varphi, \neg\varphi\} \models \psi$

An inconsistent (i.e. contradictory) set of wff entails *anything*

«*Ex absurdo sequitur quodlibet*»

